CanonicalForms

Abstract

We formulate, in the language of set theory, the concepts of invariant and canonical form for an equivalence relation $\sim$ on a set X. A map $f\colon X \to Y$ is called an invariant of $\sim$ if $x \sim y \implies f(x) = f(y)$, and a complete invariant if equality of images also implies the equivalence relation. A canonical form is an invariant $C\colon X \to X$ satisfying $x \sim C(x)$ for all $x \in X$; its image $C(X)$ is called the skeleton or complete system of representatives of $\sim$. We characterise canonical forms via retraction–section pairs, showing that C is a canonical form if and only if there exists a section s of the quotient map $q\colon X \to X/\sim$ such that $C = s \circ q$. An analogous characterisation holds for surjective complete invariants. We prove that every complete invariant $I\colon X \to Y$ factors uniquely through the surjective complete invariant $I'\colon X \to I(X)$, and define $s' \circ I'$ as the canonical form of $\sim$ determined by the complete invariant I, whenever a section s' of I' exists. Assuming the axiom of choice, we show that the canonical form determined by any complete invariant exists. Propositions that assume the axiom of choice are marked with $^{\mathrm{AC}}$.

Definition 1 Let X, Y be sets. Let $f\colon X \to Y$ be a map and let $\sim$ be an equivalence relation on X. We say that f is an **invariant** of $\sim$ if the following condition holds.

$$^{\forall}x, y \in X,\ (x \sim y \implies f(x) = f(y))$$

We further say that f is a **complete invariant** of $\sim$ if the following condition holds.

$$^{\forall}x, y \in X,\ (x \sim y \iff f(x) = f(y))$$

□

Example 1 Let X be a set. Let $\sim$ be an equivalence relation on X and let $q\colon X \to X/\sim$ be the quotient map. Then q is a complete invariant of $\sim$. □

Definition 2 Let X be sets. Let $C\colon X \to X$ be an invariant of an equivalence relation $\sim$ on X. We say that C is a **canonical form** of $\sim$ if the following condition holds.

$$^{\forall}x \in X,\ x \sim C(x)$$

We call the image $C(X)$ of X under C the **skeleton** or the **complete system of representatives** of $\sim$. □

Proposition 1 Let X be sets. Let $C\colon X \to X$ be a canonical form of an equivalence relation $\sim$ on X. Then $C = C \circ C$. $\square$

Proof Let $x \in X$. Since $x \sim C(x)$, we have $C(x) = C(C(x))$. Therefore $C = C \circ C$. $\blacksquare$

Definition 3 Let X, Y be sets and let $r\colon X \to Y$ and $s\colon Y \to X$ be maps. We call the pair (r, s) a **retraction–section pair** if the following condition holds. We call r the **retraction** of s, and s the **section** of r.

$$r \circ s = \mathrm{id}_Y$$

$\square$

Lemma 1 Let X be a set and let $q\colon X \to X/\sim$ be the quotient map of an equivalence relation $\sim$ on X. For a map $C\colon X \to X$, the following conditions (1) and (2) are equivalent.

(1) C is a canonical form of $\sim$.

(2) There exists a section $s\colon X/\sim \to X$ of $q\colon X \to X/\sim$ such that $C = s \circ q$.

$\square$

Proof (1) $\Rightarrow$ (2): Assume C is a canonical form of $\sim$. For each $y \in X/\sim$, since q is surjective there exists $x \in X$ such that $y = q(x)$. Set $s(y) = C(x)$. For any $x' \in X$ with $q(x) = q(x')$, we have $x \sim x'$, so $C(x) = C(x')$. Thus $s\colon X/\sim \to X$ is well-defined. For any $x \in X$, $C(x) = s(q(x))$, so $C = s \circ q$. For any $x \in X$, $(q \circ s)(q(x)) = q(s(q(x))) = q(C(x))$, and since $x \sim C(x)$ we have $q(C(x)) = q(x)$. Therefore $q \circ s = \mathrm{id}_{X/\sim}$.

(2) $\Rightarrow$ (1): Assume there exists a section $s\colon X/\sim \to X$ of $q\colon X \to X/\sim$ such that $C = s \circ q$. For $x, x' \in X$, if $x \sim x'$, then $C(x) = (s \circ q)(x) = s(q(x)) = s(q(x')) = (s \circ q)(x') = C(x')$, so C is an invariant of $\sim$. For any $x \in X$, $q(x) = \mathrm{id}_{X/\sim}(q(x)) = (q \circ s)(q(x)) = q((s \circ q)(x)) = q(C(x))$, so $x \sim C(x)$. Therefore C is a canonical form of $\sim$. $\blacksquare$

Proposition 2 Let X, Y be sets and let $I\colon X \to Y$ be a surjective complete invariant of an equivalence relation $\sim$ on X. For a map $C\colon X \to X$, the following conditions (1) and (2) are equivalent.

(1) C is a canonical form of $\sim$.

(2) There exists a section $s'\colon Y \to X$ of $I\colon X \to Y$ such that $C = s' \circ I$. $\square$

Proof By Lemma 1, it suffices to show that the existence of a section $s\colon X/\sim \to X$ of $q\colon X \to X/\sim$ with $C = s \circ q$ is equivalent to (2). Define $\bar{I}\colon X/\sim \to Y$ by $\bar{I}(q(x)) = I(x)$ for all $x \in X$. When $x \sim x'$, since I is an invariant we have $I(x) = I(x')$, so $\bar{I}$ is well-defined. For any $x, x' \in X$, if $I(x) = I(x')$ then, since I is a complete invariant, $x \sim x'$, hence $q(x) = q(x')$, so $\bar{I}$ is injective. For any $y \in Y$, since $I\colon X \to Y$ is surjective there exists $x \in X$ with $I(x) = y$, giving $\bar{I}(q(x)) = y$. Therefore $\bar{I}$ is bijective and hence invertible.

Suppose there exists a section s of q with $C = s \circ q$. Set $s' = s \circ \bar{I}^{-1}$. Since $q = \bar{I}^{-1} \circ I$, we have $C = s \circ \bar{I}^{-1} \circ I = s' \circ I$. Moreover, $I \circ s' = (\bar{I} \circ q) \circ (s \circ \bar{I}^{-1}) = \bar{I} \circ (q \circ s) \circ \bar{I}^{-1} = \bar{I} \circ \mathrm{id}_{X/\sim} \circ \bar{I}^{-1} = \mathrm{id}_Y$, so s' is a section of I.

Conversely, suppose (2) holds. Set $s = s' \circ \bar{I}$. Then $s \circ q = s' \circ \bar{I} \circ q = s' \circ I = C$. Since $I \circ s' = \mathrm{id}_Y$, we have $I \circ s' \circ I = I$, and therefore $I \circ C = I$. Since $I = \bar{I} \circ q$, it follows that $\bar{I} \circ q \circ C = \bar{I} \circ q$, and by injectivity of $\bar{I}$ we obtain $q \circ C = q$. For any $y \in X/\sim$ there exists $x \in X$ such that $y = q(x)$. Then

$$(q \circ s)(y) = (q \circ s' \circ \bar{I})(y) = (q \circ s' \circ \bar{I} \circ q)(x) = (q \circ s' \circ I)(x) = (q \circ C)(x) = q(x) = y,$$

so $q \circ s = \mathrm{id}_{X/\sim}$. ■

Proposition 3 Let X, Y be sets and let $I\colon X \to Y$ be a complete invariant of an equivalence relation $\sim$ on X. The unique map $I'\colon X \to I(X)$ satisfying $I = i \circ I'$ with respect to the inclusion $i\colon I(X) \to Y$ is surjective and is a complete invariant of $\sim$. □

Proof Define $I'\colon X \to I(X)$ by $I'(x) = I(x)$ for all $x \in X$. Since $I = i \circ I'$ and $I'(X) = I(X)$, I' is surjective. For any $x, x' \in X$, $x \sim x' \iff I(x) = I(x') \iff I'(x) = I'(x')$, so I' is a complete invariant of $\sim$. If $I''\colon X \to I(X)$ satisfies the condition, then $I'(x) = I(x) = I''(x)$ for all $x \in X$, so I' is unique. ■

Definition 4 Let X, Y be sets and let $I\colon X \to Y$ be a complete invariant of an equivalence relation $\sim$ on X. When a section $s'\colon I(X) \to X$ of the unique map $I'\colon X \to I(X)$ satisfying $I = i \circ I'$ with respect to the inclusion $i\colon I(X) \to Y$ exists, we call the canonical form $s' \circ I'$ of $\sim$ the **canonical form of $\sim$ determined by the complete invariant I**. □

Lemma 2$^{\mathrm{AC}}$ Let X, Y be sets and let $f\colon X \to Y$ be a surjection. Then there exists a map $g\colon Y \to X$ such that $f \circ g = \mathrm{id}_Y$. □

Proof Since $f\colon X \to Y$ is surjective, $f^{-1}(y) \neq \emptyset$ for every $y \in Y$. Therefore, by the axiom of choice, $\prod_{y \in Y} f^{-1}(y) \neq \emptyset$, so there exists $g \in \prod_{y \in Y} f^{-1}(y)$. This map $g\colon Y \to X$ satisfies $f \circ g = \mathrm{id}_Y$. ■

Proposition 4$^{\mathrm{AC}}$ Let X, Y be sets and let $\sim$ be an equivalence relation on X. Then the canonical form of $\sim$ determined by the complete invariant I exists. □

Proof Let $q\colon X \to X/\sim$ be the quotient map. Since q is surjective, by Lemma 2 there exists a map $s\colon X/\sim \to X$ satisfying $q \circ s = \mathrm{id}_{X/\sim}$. By Lemma 1, $s \circ q$ is a canonical form of $\sim$, and by Proposition 2, taking I' to be the surjection determined by I, there exists a section $s'\colon I(X) \to X$ of I' such that $s \circ q = s' \circ I'$. Then $s' \circ I'$ is the canonical form of $\sim$ determined by the complete invariant I. ■